

Xinru (Raya) Cheng

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EDUCATION

Peking University

Beijing, China

Master of Software Engineering

Sep 2024 – Jul 2027

- Major Courses: Algorithm analysis and design, Game engines and game development, Natural language processing, Machine learning, Artificial intelligence practices, Software implementation technology, etc.
- Rank: 24/279 (Top 10%)

Beijing University of Chemical Technology

Beijing, China

Bachelor of Digital Media Art

Sep 2020 – Jul 2024

- Major Courses: Programming Foundation, Advanced Mathematics, Game Planning and Development, Advanced Practicum, Virtual Reality Technology, Three Dimensional Animation Design and Production, Graphic Image Processing, Practicum in Applied Software, Graphics of Engineering, etc.
- Rank: 1/30 (Direct Admission to Peking University)

ACADEMIC PAPERS

Full Papers

- **Cheng, X.**, Zhang, Y., Wei, Y., Jiang, R., Hansen, P., Liang, H. N. Effects of Spatial Constraint on Bare-Hand Logographic Character Handwriting Performance in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J] (Accepted).
- Wei, Y., **Cheng, X.**, Shi, R., Liang, H. N. Reorienting with the Bare Hand: Gesture-Based Techniques for Orientation-Enabled Teleportation in Virtual Reality. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J].
- Gao, Y., **Cheng, X.**, Li, Y., Xu, D., Sun, D., Hu, Y. ISDG-Net: Efficient RGB-Infrared Object Detection for Remote Sensing Imagery. In *Remote Sensing* [J].
- Zhang, Y., **Cheng, X.**, Yan, J. AI-Echo: Exploring Technological Alienation Through an Interactive Art Experience. In *International Conference on Human-Computer Interaction (HCII'25)* [C].
- Jiang, R., Zhang, S., Zhao, Y., **Cheng, X.**. The Application of Serious Games in Inclusive Education. In *Game Industry Blue Book: China Game Industry Development Report (2025)*. Beijing: Social Sciences Academic Press [BC].

Short Papers

- **Cheng, X.**, Zhang, Y., Qiao, Y., Hansen, P., Jiang, R. Negative Danmaku as Social Warning Cues in Video Credibility Assessment. In *European Conference on Computer-Supported Cooperative Work (ECSCW'26)* (Poster)[C].
- Zhang, Z., Zhang, Y., **Cheng, X.***. Private Non-Visual Addressivity Cues for Blind and Low Vision Users in Mixed-Ability Social VR. In *Conference on Computer-Supported Cooperative Work (CSCW'26)* (Poster) [C] (Accepted).
- **Cheng, X.**, Wei, Y., Liang, H. N. Toward AI-Mediated Candidate Delivery for Chinese Pinyin Input in Virtual Reality. In *1st International Workshop on AI-Mediated Heads-Up Interaction in Wearable Mixed Reality (AHIMR'26)* & In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR'26)* (Workshop) [C] (Accepted).
- Li, Z., Huang, K., **Cheng, X.**, Tang, H., Li, H., Zhao, S., An, P. FlowWalk: AR Guided Outdoor Mindful Walking with Photo Capture Based Nature Awareness. In *ACM International Joint Conference on Pervasive and Ubiquitous Computing / International Symposium on Wearable Computers*

(*UbiComp/ISWC'26*) (Poster) [C] (Accepted).

- Li, Z., Huang, K., Tang, H., **Cheng, X.**, Li, H., Zhao, S., An, P. Demonstrating FlowWalk: A Smart Glasses System for Outdoor Mindful Walking and Connection with Nature. In *ACM International Joint Conference on Pervasive and Ubiquitous Computing / International Symposium on Wearable Computers (UbiComp/ISWC'26)* (Demonstration) [C] (Accepted).

* Corresponding author.

PATENTS

- Zhang, P., Li, W., Tang, Z., Wang, Z., Jin, F., **Cheng, X.** *An AI Agent System and Collaborative Knowledge Construction Method*. Chinese Invention Patent, ZL 2025 1 1737406.2, Granted May 19, 2026.

RESEARCH EXPERIENCE

Research Assistant, Computational Media and Arts Thrust, Hong Kong University of Science and Technology (Guangzhou) Jul 2025 – Sep 2026

- Supervisor: Hai-Ning Liang
- Finished experimental design, development, and user studies for three parallel projects on VR resolution, reorientation techniques, and handwriting interaction; contributed to three manuscripts accepted/submitted to IEEE TVCG/VR/ISMAR.

Research Intern, Department of Computer Science and Engineering, Hong Kong University of Science and Technology Dec 2025 – Sep 2026

- Supervisor: Xiaojuan Ma
- Participated in the conceptualization and research design of specific tasks such as the human-computer interaction research direction related to speculative artificial products, human-machine collaboration, and the support of large language models.

Visiting Student, Department of Information Management, Peking University Feb 2025 – Jun 2025

- Supervisor: Pengyi Zhang / Preben Hansen
- Participated in interaction design and front-end development for two AI agent-based educational systems supporting collaborative lesson planning, interactive learning, and knowledge construction; contributed to an NSFC-funded project and a granted Chinese invention patent.

Research Intern, National Virtual Simulation Experimental Teaching Center for Earth Sciences, Peking University Dec 2024 – Jun 2025

- Supervisor: Yanjun Guo
- Conducted research on immersive virtual environments for marine science education, investigating how sense of presence and natural connectedness influence learners' pro-environmental behavioral intentions.

Research Intern, Future Lab - Space and Media Group, Tsinghua University May 2023 – Jul 2023

- Supervisor: Keyang Tang
- Assisted in the development and design of the Digital Luoyang human-computer interaction system.

INDUSTRIAL EXPERIENCE

Investment Intern, Alpha Startups Aug 2025 – Nov 2025

- Conducted pre-investment research, focusing on next-generation interaction technologies and products.
- Participated in all stages of the investment process, including company analysis and market research.

Research Intern, MiraclePlus (YC China) Oct 2024 – Nov 2024

- Responsible for communicating with entrepreneurs and planning Demo Day events.

Product Manager Intern, Kuaishou Technology Apr 2024 – Aug 2024

- Worked on product development for the e-commerce C-end product department.

Product Manager Intern, DiDi Global Nov 2023 – Apr 2024

- Worked on growth product initiatives for the Didi platform, Chitu, and Didi Alliance.

Interaction Design Intern, Meituan

Jul 2023 – Nov 2023

- Responsible for interaction design and AI capability building in Meituan Cuisine.
- Received a full-time offer after internship.

ACADEMIC SERVICES

- Teaching Assistant - Game Development Game Engines, Peking University Sep 2024 – Jan 2025
- Teaching Assistant - Artificial Intelligence Practice, Peking University Mar 2025 – Jul 2025
- Reviewer - ISMAR 2025/2026
- Reviewer - CHI 2026
- Reviewer - C&C 2026
- Reviewer - UIST 2026
- Reviewer - CHI PLAY 2026

HONORS & AWARDS

- **National Scholarship (top 1%)**, 2022.
- Shenzhen Stock Exchange Scholarship, 2025 (one of 17 recipients university-wide).
- Outstanding Student Award, Peking University, 2025.
- People's Scholarship (top 5%), 2021,2022,2023.
- Best Student Short Film, Hollywood Independent Film Festival, 2022.
- Third Prize, National College Student Advertising Art Competition, 2022.

SKILLS

- **Programming & Development:** Python, C#, C++, Blueprint, JavaScript, HTML, CSS
- **XR & Creative Technologies:** Unity, Unreal Engine, Processing, TouchDesigner, Kinect, Leap Motion, Arduino
- **3D Modeling & Rendering:** Blender, 3DS MAX, Substance Painter, Zbrush, Octane Render, Marmoset Toolbag, PBR Material Workflow
- **Design & Media Software:** Figma, Adobe Photoshop, After Effects, Animate, Premiere Pro, Illustrator
- **Statistical Analysis:** Descriptive Statistics, Hypothesis Testing (t-test, Chi-Square Test, ANOVA), Regression Analysis, Non-Parametric Tests
- **Data Analysis:** SPSS, Python (pandas, statsmodels), Excel
- **Hobbies:** Pilates, Diving, FPS Games, Piano(Grade 10 of Wuhan Conservatory of Music)